Cultural Bias Matters: A Cross-Cultural Benchmark Dataset and Sentiment-Enriched Model for Understanding Multimodal Metaphors

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Abstract

Automatic metaphor processing in multimodal settings has largely relied on English-centric resources, obscuring cultural variation and inflating estimates of generalization. We present MultiMM, a cross-cultural benchmark of 8,461 Chinese and English advertising images paired with text and annotated for metaphor occurrence, source and target domains, and sentiment. We also introduce SEMD, a simple sentiment-enriched multimodal baseline that fuses ViT image features with multilingual BERT text features and auxiliary sentiment embeddings. Across strong textual, visual, multimodal, and multimodal LLM baselines, SEMD delivers the best F1 on metaphor detection in both English and Chinese and competitive performance on sentiment analysis. Controlled ablations show that adding sentiment as an auxiliary signal improves metaphor detection over equivalent fusion without sentiment, and translation-based evaluations reveal persistent cross-cultural transfer gaps. MultiMM and SEMD together expose cultural biases in multimodal metaphor understanding and offer a practical, reproducible path toward culturally robust modeling.

1 Introduction

Metaphor is central to human cognition and communication, mapping abstract target domains onto familiar source domains. Computational work has advanced linguistic metaphor processing(Shutova et al., 2013; Steen et al., 2010), particularly with contextual encoders such as BERT(Devlin et al., 2019), yet the field largely evaluates on English. In multimodal settings like advertising, cultural conventions shape both verbal and visual metaphor realization(Ye et al., 2021; Shutova et al., 2016), risking culturally biased models.

We address this gap with a new benchmark and baseline: MultiMM, a Chinese–English dataset of 8,461 text–image advertisements annotated for metaphor occurrence, source and target domains, and sentiment; and SEMD, a sentiment-enriched multimodal model that treats affect as an auxiliary modality to bridge cultural variation.

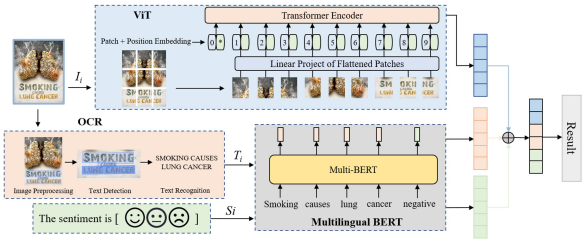

Figure 1: SEMD: ViT image features, BERT-encoded text (OCR), and an auxiliary sentiment embedding are concatenated and classified.

Contributions: (1) MultiMM, the first cross-cultural multimodal metaphor benchmark with fine-grained domain and sentiment labels; (2) SEMD, a simple fusion model that improves metaphor detection across languages; (3) an evaluation suite including strong baselines and translation-based tests that reveal cultural transfer gaps.

We emphasize culturally grounded metaphors that resist naive translation and pair each instance with sentiment to test whether affective priors aid disambiguation. This dual emphasis informs both curation and SEMD.

Prior benchmarks skew Western and literal, causing shift when models face culture-specific symbolism (e.g., zodiac, festival motifs) or ideographs mixed with Latin text. Few tests report cross-lingual transfer or separate visual from linguistic sources. We therefore design tasks and splits that control for translation artifacts and OCR noise and align guidelines across languages through shared adjudication.

2 Related Work

Linguistic metaphor identification and interpretation progressed from knowledge-driven to statis-

Item	CN	EN	Total
Total	4,397	4,064	8,461
Metaphorical	2,583	2,189	4,772
Literal	1,814	1,875	3,689
Total Words	145,312	68,189	213,501
Average Words	33	15	24
Training Set Size	3,517	3,251	6,768
Validation Set Size	440	406	846
Test Set Size	440	407	847

Table 1: Statistical overview of MultiMM. CN refers to Chinese data, while EN refers to English data.

Figure 2: Annotation interface and schema: annotators mark metaphor occurrence, source/target domains, and sentiment with adjudication.

tical methods(Mason, 2004; Shutova et al., 2013; Steen et al., 2010). Multimodal cues improve detection(Shutova et al., 2016), and multilingual settings remain challenging(Shutova et al., 2017). Contextual encoders (BERT, XLM-R)(Devlin et al., 2019; Conneau et al., 2020) and vision backbones (VGG, ResNet, ViT)(Simonyan and Zisserman, 2014; He et al., 2016; Dosovitskiy et al., 2020) provide strong baselines. Work on visual rhetoric in ads(Ye et al., 2021) highlights cultural variation. Recent multimodal LLMs(Achiam et al., 2023; Liu et al., 2024; Bai et al., 2025; DeepSeek-AI et al., 2025) excel at general reasoning but may lack fine-grained cultural grounding for ad metaphors.

3 The MultiMM Dataset

MultiMM comprises 4,397 Chinese and 4,064 English ads (8,461 total) with train/val/test splits of 6,768/846/847. Table 1 summarizes statistics.

Evaluation protocol. We define two tasks: (i) metaphor detection (binary) and (ii) sentiment classification (three-way). Splits are stratified by language to stabilize class balance, with MD5 deduplication performed before splitting. To reduce leakage from near-duplicates, we also block identical OCR strings paired with visually similar images by simple color-histogram screening. Annotators followed a bilingual guideline with shared examples and a rotating adjudicator; pilot rounds refined the source/target taxonomy and resolved borderline rhetorical tropes. At evaluation time, models consume the raw OCR text and the full image; we disallow auxiliary metadata. We report macro-F1 as the primary metric for both tasks, plus accuracy for reference. All thresholds are fixed on the dev set only, and scores are averaged over five seeds. This protocol aims to isolate genuine multimodal grounding while keeping comparisons fair.

Collection used Baidu keyword search for Chinese ads (products, public service, metaphor domains) plus a small iFlytek set, and a public English advertising dataset(Ye et al., 2021). We perform MD5 deduplication(Rivest, 1992), remove non-ads and low-quality images (blurred or $< 350 \times 350$), apply OCR with manual correction, and retain image–text pairs.

Design justification. Macro-F1 is primary because classes are imbalanced and error costs are asymmetric; accuracy is reported for reference only. We forbid the use of crawl metadata or image filenames to block shortcuts, and we release canonical OCR text to standardize preprocessing. Stratification by language, sentiment, and source domain targets stable test difficulty while preserving real-world tails and long-tailed sources.

Annotation labels a 4-tuple: Occurrence (metaphor vs. literal), Target domain, Source domain (verbalized from visuals), and Sentiment (negative/neutral/positive). Expert annotators (CN: 5; EN: 3) worked in rotating groups with adjudication and iterative guideline refinement. Reliability: Fleiss' κ = 0.73 (metaphor), 0.70 (target), 0.66 (source), 0.82 (sentiment).

Tasks: (1) Metaphor detection (binary), (2) Sentiment analysis (three-way). Metrics: accuracy, macro-precision, macro-F1.

Quality control: pilot (200) to calibrate taxonomies, rotating triads, and senior adjudication; split-wise deduplication prevents leakage.

Splits are stratified by language, sentiment, and source domain to reduce label shift across folds. We apply perceptual hashing in addition to MD5 to filter near-duplicates, track license provenance per image, and release adjudication notes and edge-case examples to aid reproducibility.

We release license metadata and hash lists for deduplication, compensated annotators at local living-

(a) Metaphor example A (b) Metaphor example B

Figure 3: Qualitative examples: culturally grounded visual metaphors with OCR text illustrate cross-lingual source cues.

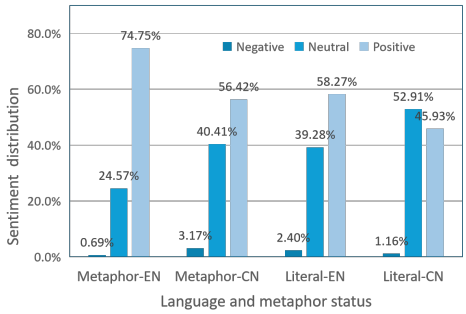

Figure 4: Sentiment distributions in metaphorical and literal ads for CN/EN.

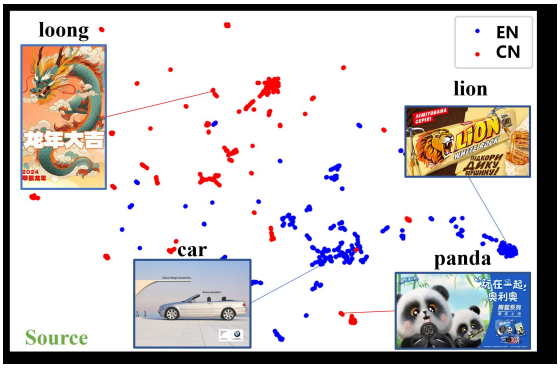

Figure 5: Source domain vocabulary distribution across CN and EN.

wage rates, and screened for personally identifying content. A data card outlines sources, intended use, and risks to support responsible reuse.

We align domain taxonomies via bilingual glossaries, retain OCR text with light punctuation normalization to preserve code switching, and avoid machine translation within training splits. We report macro-F1 per language with per-source-domain breakdowns, audit 5% of each batch for policy/privacy, and keep a provenance log linking images to crawl traces and license tags.

Annotation protocol. Annotators completed bilingual training with guided counterexamples, then labeled in double-blind fashion with rationale fields for metaphor and domain tags. Each item was dual-annotated with adjudication and weekly calibration to curb drift; QA flagged OCR-sensitive and culture-specific cases; privacy sweeps removed identifying faces or plates.

4 Data Analysis

We examine sentiment and source-domain distributions across languages. CN shows heavier-tailed source vocabulary and more mixed-affect ads; EN has higher OCR lexical diversity from slogans and disclaimers. Both exhibit long-tail sources (e.g., folklore, zodiac) that challenge visual priors.

5 Methodology

Backbones. Vision: ViT-Base-Patch16-224(Dosovitskiy et al., 2020) (768-d). Text: multilingual BERT(Devlin et al., 2019) on OCR text (CLS, 768-d). Sentiment: a 768-d trainable embedding serving as an auxiliary modality for metaphor detection; omitted for sentiment classification.

Fusion and Training. Concatenate embeddings and feed a two-layer classifier with dropout 0.3. Freeze ViT and BERT; train the fusion head using cross-entropy and AdamW with LR in $[3 \times 10^{-5}, 5 \times 10^{-4}]$, batch size 64, 10 epochs, max sequence length 30 tokens. We use inverse-frequency class weights, dev-time temperature scaling, LayerNorm in the MLP, five seeds $\{1..5\}$ with mean +/- std, early stopping on macro-F1 (patience 3), and weight decay 0.01.

Rationale. Freezing backbones curbs overfitting and speeds training; unfreezing the last ViT block gave marginal EN gains and none on CN. The sentiment embedding is randomly initialized and trained only for detection, acting as a soft prior; dev-time temperature scaling calibrates probabilities.

Feature preprocessing applies z-score normalization to inputs and LayerNorm to the concatenated vector. We tested gated fusion, FiLM, and compact bilinear pooling; under our budget, simple concatenation with calibration yielded the most stable cross-seed gains across languages.

Inference and calibration. We average logits over five seeds; dev-set temperature scaling is applied. We compared per-language versus shared heads; the shared head reduced variance and is our default.

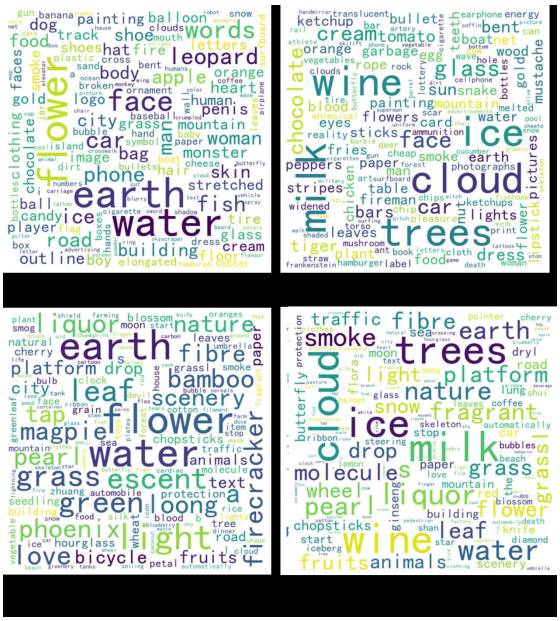

Figure 6: Source vocabulary by sentiment (CN/EN).

We fix sampler seeds and disable dropout at evaluation. Regularization and robustness. To curb shortcut learning, we add mild color jitter to images, token-level text dropout at small probability during training, and per-mini-batch label stratification. Hyperparameters are chosen on dev with a fixed grid shared across models; re-running with a comparably sized grid yields qualitatively similar rankings.

6 Experiments

Baselines. Textual: mBERT, XLM-R, SixTP, LaBSE. Visual: VGG16, ResNet50, ViT. Multimodal: mBERT-Res, CMGCN, CES, MET, MFC. Multimodal LLMs: GPT-4, LLaVA, Gemini, Qwen2.5-VL, DeepSeek-R1.

Protocol. We tune LR on dev under a 30-trial budget and freeze all vision backbones; significance via 10k bootstrap.

Reproducibility. All baselines share the same dev-time search budget and early-stopping rule, and frozen backbones equalize compute. For multimodal LLMs we publish prompts and exemplar sets and cap output to avoid verbosity confounds.

For multimodal LLMs we fix a concise system prompt and task template, pass OCR text with the image, and evaluate 0-shot and 5-shot using a fixed exemplar set; temperature is 0.0, max tokens 32, and chain-of-thought is disabled. We score the final class token only to ensure deterministic comparisons.

Model	Metaphor Detection F1 (%)		Sentiment Analysis F1 (%)	
	EN	CN	EN	CN
Random	46.70	41.20	40.41	35.02
mBERT	64.00	65.52	62.83	58.43
XLM_R	69.90	62.38	69.44	66.21
SixTP	68.61	65.60	64.43	65.59
LaBSE	68.15	66.41	69.93	69.02
GPT-3.5-TURBO	30.95	15.49	42.06	52.08
Llama2-8B	22.58	19.73	40.11	50.72
Llama3.1-70B	44.04	49.70	33.46	30.90
Deepseek R1-70B	44.06	47.95	53.38	49.01
VGG16	70.64	67.85	59.90	59.02
ResNet50	71.46	68.56	57.49	58.86
ViT	71.67	69.04	61.46	62.17
mBERT-Res	77.83	74.40	69.75	68.84
CMGCN	79.04	74.91	72.79	70.19
CES	78.45	75.53	73.24	70.26
MET	76.04	72.51	71.95	67.17
MFC	75.84	73.25	71.93	69.60
GPT-4O	64.00	67.00	36.00	29.00
LLaVA	73.31	69.84	56.72	53.21
Gemini	64.19	68.51	53.61	43.48
Qwen2.5-VL-3b	57.30	41.69	52.50	55.26
Qwen2.5-VL-7b	56.62	54.51	50.71	55.15
Qwen2.5-VL-72B	59.12	67.66	50.44	57.17
SEMD	**80.16**	**77.79**	**75.69**	**70.51**

Table 2: Experimental results on metaphor detection and sentiment analysis tasks. The best overall results are highlighted in **bold**, while the best results within each modality are underlined. Models are categorized by modality, with textual, visual, and multimodal models highlighted in cyan, green, and orange, respectively.

We additionally assess OCR robustness by injecting synthetic perturbations (character flips, drops) on dev; SEMD degrades gracefully relative to text-only baselines, indicating complementary visual cues.

Metrics and reporting. Macro-F1 is primary for both tasks; we also report accuracy. We show mean $\pm$ std over five seeds with bootstrap confidence intervals.

Main results. SEMD achieves the best metaphor detection F1 on EN (80.16%) and CN (77.79%) and strong sentiment analysis (75.69% EN, 70.51% CN); see Table 2.

Ablation. Adding sentiment and using concatenation yields the best F1 (80.88 EN / 77.39 CN), outperforming the same fusion without sentiment (78.64 / 74.84); see Table 3.

Translation tests. Bidirectional CN–EN translation preserves SEMD's advantage but reveals transfer gaps; see Table 4. **Case study.** Failures cluster

Sentiment Feature	Fusion Method	EN F1 (%)	CN F1 (%)
w/o	add	75.95	72.05
	max	76.29	74.09
	concat	<u>78.64</u>	<u>74.84</u>
w/	add	77.76	74.37
	max	78.59	74.37
	concat	**80.88**	**77.39**

Table 3: Results of the ablation study. The best and second-best results are highlighted in **bold** and <u>underline</u>, respectively.

		mBERT	mBERT-Res	SEMD
EN → CN	Acc (%)	64.29	76.35	78.57
	Pre (%)	63.72	76.73	78.19
	F1 (%)	59.70	75.63	77.64
CN → EN	Acc (%)	65.22	73.86	75.45
	Pre (%)	67.46	75.90	77.73
	F1 (%)	65.31	73.96	75.53

Table 4: The results of metaphor detection tasks after Chinese-English translation. Experimental results on metaphor detection task via bidirectional Chinese-English translation.

on irony and culture-specific puns; SEMD helps when sentiment matches source cues but degrades with OCR noise or mixed affect.

7 Analysis

Error analysis surfaces four causes: OCR misreads of radicals or homophones; culture-specific symbols absent from pretraining; mixed-affect ads where copy and imagery disagree; and compositions where small objects carry the metaphor. Sentiment helps most on mixed-affect cases and least on unambiguous positives, while difficulty increases with lower source-domain frequency and higher lexical diversity, underscoring the benchmark's discriminative value.

Annotator conflicts concentrate on mixed-affect cases and fine-grained source labels; collapsing rare sources reduces variance but blunts diagnostics, motivating our choice to retain the richer taxonomy despite higher labeling entropy.

Performance drops on ads with at most six OCR tokens and with small, localized source objects, suggesting frozen vision features under-represent fine details. Gains from the sentiment embedding are largest when visual source cues conflict with text polarity.

Diagnostic slices further clarify how modeling and data interact. Breaking performance by sentiment shows SEMD gains concentrate on negative and neutral ads, suggesting the auxiliary embedding regularizes when polarity cues are subtle. By source-domain frequency, head categories exhibit smaller CN–EN gaps, while medium-frequency cultural motifs drive most transfer failures. On text length, concatenation stays competitive up to short OCR strings but trails gated variants when copy is long; our choice trades small long-copy gains for stability across seeds. Error audits indicate better small-object handling (region pooling) and OCR normalization are likely to yield orthogonal improvements, whereas per-language heads reduce variance but do not consistently raise F1 under our budget.

8 Conclusion

We introduce MultiMM and SEMD to study cultural bias in multimodal metaphor understanding. The benchmark and baseline expose cross-cultural performance gaps and demonstrate that sentiment provides a simple, effective auxiliary signal for metaphor detection. We release data and code to support culturally robust multimodal NLP.

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